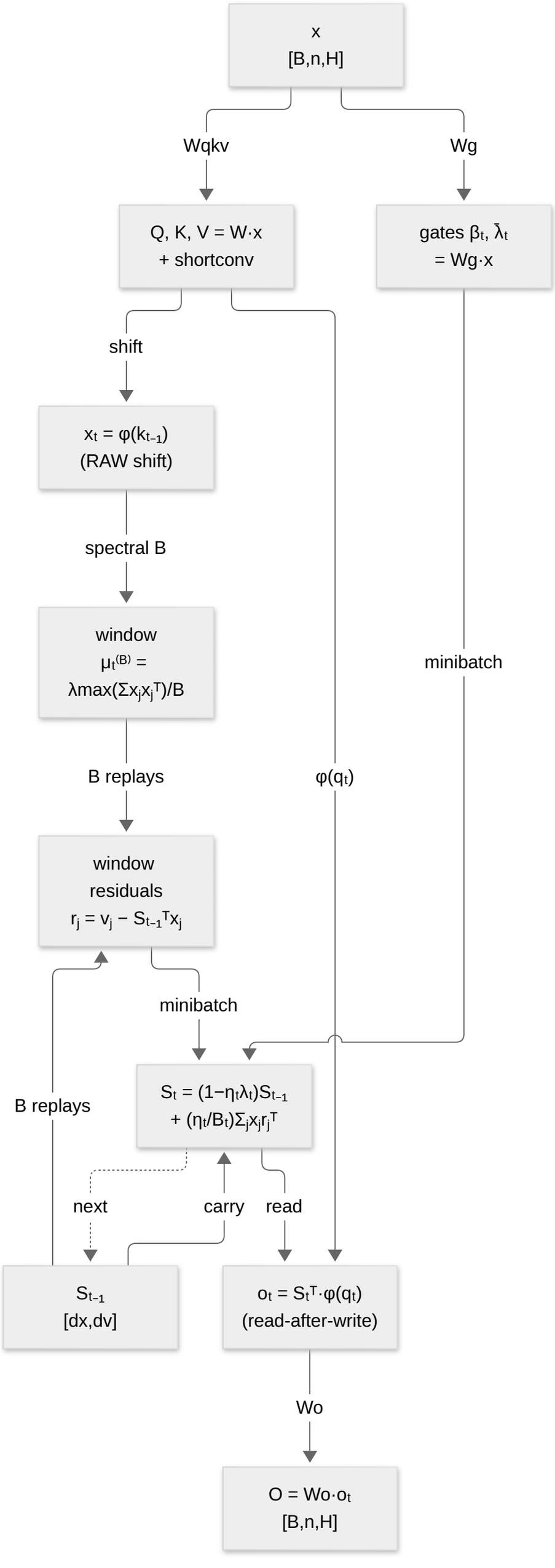

x
[B,n,H]
Wqkv
Wg
Q, K, V = W·x
+ shortconv
gates βt, λ̄t
= Wg·x
shift
xt = φ(kt−1)
(RAW shift)
spectral B
window
μt(B) =
λmax(ΣxjxjT)/B
B replays
φ(qt)
minibatch
window
residuals
rj = vj − St−1Txj
minibatch
B replays
St = (1−ηtλt)St−1
+ (ηt/Bt)Σjxjrj T
next
carry
read
St−1
[dx,dv]
ot = StT·φ(qt)
(read-after-write)
Wo
O = Wo·ot
[B,n,H]